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Magnetic sensor array

Abstract

A magnetic sensor array includes a plurality of magnetic sensors. The magnetic sensor includes a sensor housing body housing the sensor chip and first and second connection parts formed on the sensor housing body. The first connection part formed on one of adjacent magnetic sensors and the second connection part formed on the other one thereof are axially supported on each other to achieve mutual connection between the plurality of magnetic sensors. Thus, it is possible to vary the relative angle between two adjacent magnetic sensors with the first and second connection parts as an axis. This allows the plurality of magnetic sensors to be pressed perpendicularly against even a curved surface of an object to be measured, and the pressed state can be easily maintained.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This application is the U.S. National Phase under 35 U.S.C. § 371 of International Application No. PCT/JP2021/020252, filed on May 27, 2021, which claims the benefit of Japanese Patent Application No. 2020-104819, filed on Jun. 17, 2020, the entire contents of each are hereby incorporated by reference.

TECHNICAL FIELD

(2) The present invention relates to a magnetic sensor array including a plurality of magnetic sensors and, more particularly, to a magnetic sensor array in which the angle between the plurality of magnetic sensors is variable.

BACKGROUND ART

(3) Patent Document 1 discloses in FIG. 2 thereof a magnetic sensor array in which a plurality of magnetic sensors can be slid independently in one direction. Patent Document 1 further discloses in FIG. 14 thereof a magnetic sensor array in which a plurality of magnetic sensors are held by a holding part made of a flexible material. In these magnetic sensor arrays, a plurality of magnetic sensors can be brought close to even a curved surface of an object to be measured.

CITATION LIST

Patent Document

(4) [Patent Document 1] International Publication WO 2018/02582.

SUMMARY OF THE INVENTION

Problem to be Solved by the Invention

(5) However, in the magnetic sensor array illustrated in FIG. 2 of Patent Document 1, the magnetic sensors can be slid only in one direction and thus cannot be pressed perpendicularly against the surface of an object to be measured when the surface is a curved one. On the other hand, in the magnetic sensor array illustrated in FIG. 14 of Patent Document 1, although the magnetic sensors can be pressed perpendicularly against even a curved surface of an object to be measured, it is difficult to maintain a state where the magnetic sensors are pressed perpendicularly against the surface of an object to be measured due to a flexible holding part.

(6) It is therefore an object of the present invention to provide a magnetic sensor array in which magnetic sensors constituting the magnetic sensor array can be pressed perpendicularly against even a curved surface of an object to be measured, and the pressed state can be easily maintained.

Means for Solving the Problem

(7) A magnetic sensor array according to the present invention is a magnetic sensor array including a plurality of magnetic sensors. The magnetic sensors each include a sensor chip, a sensor housing body housing the sensor chip, and first and second connection parts formed on the sensor housing body. The first connection part formed on one of adjacent magnetic sensors and the second connection part formed on the other one thereof are axially supported on each other to achieve mutual connection between the plurality of magnetic sensors.

(8) According to the present invention, it is possible to vary the relative angle between two adjacent magnetic sensors with the first and second connection parts as an axis. This allows the plurality of magnetic sensors to be pressed perpendicularly against even a curved surface of an object to be measured, and the pressed state can be easily maintained.

(9) In the present invention, the sensor housing body may have a sensor head perpendicular to the sensitivity axis direction of the magnetic sensor and first and second side surfaces parallel to the sensitivity axis direction of the magnetic sensor and positioned on mutually opposite sides, the first connection part may be formed on the first side surface at a portion in a vicinity of the sensor head, and the second connection part may be formed on the second side surface at a portion in a vicinity of the sensor head. This arrangement makes it possible to deform the plurality of magnetic sensors connected in one direction into a fan shape.

(10) In the present invention, the plurality of magnetic sensors may constitute a plurality of groups

including first and second groups, at least one of the magnetic sensors belonging to the first group may further include a third connection part formed on the sensor housing body, at least one of the magnetic sensors belonging to the second group may further include a fourth connection part formed on the sensor housing body, and the first and second groups may be mutually connected with the third and fourth connection parts axially supported on each other. This makes it possible to vary the relative angle between the plurality of magnetic sensors belonging to the same group with the first and second connection parts as an axis and to vary the relative angle between the plurality of magnetic sensors belonging to different groups with the third and fourth connection parts as an axis.

(11) In the present invention, the sensor housing body may further have third and fourth side surfaces parallel to the sensitivity axis direction of the magnetic sensor and perpendicular to the first and second side surfaces and are positioned on mutually opposite sides, the third connection part may be formed on the third side surface at a portion in a vicinity of the sensor head, and the fourth connection part may be formed on the fourth side surface at a portion in a vicinity of the sensor head. This makes it possible to deform a measurement surface constituted by the plurality of sensor heads into a spherical shape.

(12) In the present invention, a rotary shaft axially supporting the first and second connection parts may be made of a non-magnetic material. This prevents the rotary axis from causing magnetic noise.

Advantageous Effects of the Invention

(13) As described above, according to the present invention, there can be provided a magnetic sensor array in which magnetic sensors constituting the magnetic sensor array can be pressed perpendicularly against even a curved surface of an object to be measured, and the pressed state can be easily maintained.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic perspective view illustrating the outer appearance of a magnetic sensor array 1 according to a first embodiment of the present invention.

(2) FIG. 2 is a schematic view illustrating an example of the internal structure of the magnetic sensor 10.

(3) FIG. 3 is a schematic plan view of the sensor chip 22.

(4) FIG. 4 is a cross-sectional view taken along the line A-A in FIG. 3.

(5) FIG. 5 is a circuit diagram of the magnetic sensor 10.

(6) FIG. 6 is a schematic perspective view illustrating a state where the magnetic sensor array 1 is deformed.

(7) FIG. 7 is a schematic perspective view illustrating the outer appearance of a magnetic sensor array 2 according to a second embodiment of the present invention.

(8) FIG. 8 is a schematic perspective view illustrating the outer appearance of a magnetic sensor array 3 according to a third embodiment of the present invention.

(9) FIG. 9 is a schematic perspective view illustrating a state where the magnetic sensor array 3 is deformed.

MODE FOR CARRYING OUT THE INVENTION

(10) Hereinafter, preferred embodiments of the present invention will be described in detail with reference to the accompanying drawings.

(11) FIG. 1 is a schematic perspective view illustrating the outer appearance of a magnetic sensor array 1 according to a first embodiment of the present invention.

(12) As illustrated in FIG. 1, the magnetic sensor array 1 according to the first embodiment has a

configuration in which a plurality of magnetic sensors **10** each elongated in the z-direction are arranged in the x-direction. The sensitivity axis direction of each of the magnetic sensors is along the z-direction. In the example illustrated in FIG. 1, seven magnetic sensors **10** constitute the magnetic sensor array **1**; however, the number of the magnetic sensors **10** constituting the magnetic sensor array **1** is not particularly limited to a specific number.

(13) Each magnetic sensor **10** includes a sensor housing body **20**, a connection part **11** formed on a side surface **S1** of the sensor housing body **20**, and a connection part **12** formed on a side surface **S2** of the sensor housing body **20**. The side surfaces **S1** and **S2** each constitute the yz surface and are positioned on mutually opposite sides. As illustrated in FIG. 1, each sensor housing body **20** has a pair of connection parts **12**. The interval between the pair of connection parts **12** in the y-direction is substantially the same as or slightly smaller than the thickness of the connection part **11** in the y-direction, and the connection part **11** of one of the magnetic sensors **10** adjacent in the x-direction is inserted between the pair of connection parts **12** of the other one thereof. Then, a rotary shaft **15** is inserted into through holes formed in the connection parts **11** and **12**, whereby the connection parts **11** and **12** are axially supported on each other by the rotary shaft **15**. The axial direction of the rotary shaft **15** is along the y-direction and does not overlap the magnetic sensor **10**. The material constituting the rotary shaft **15** is not particularly restricted as long as it can axially rotatably support the connection parts **11** and **12** and may be constituted by a screw or an elastic body. The rotary shaft **15** is preferably made of a non-magnetic material.

(14) FIG. 2 is a schematic view illustrating an example of the internal structure of the magnetic sensor **10**.

(15) As illustrated in FIG. 2, the magnetic sensor **10** includes a sensor housing body **20** made of a non-magnetic material, a substrate **21** housed in the sensor housing body **20**, a sensor chip **22**, and a magnetism collector **23**. The sensor chip **22** and magnetism collector **23** are mounted on the substrate **21**. The magnetism collector **23** has a rod-like shape extending in the z-direction and is made of a high permeability material such as ferrite. The xy surface of the sensor housing body **20** at which one end of the magnetism collector **23** in the z-direction is positioned constitutes a sensor head **H**, and the sensor chip **22** is disposed at the other end of the magnetism collector **23** in the z-direction. With this configuration, a signal magnetic field component in the z-direction emitted from an object to be measured positioned in the vicinity of the sensor head **H** is collected by the magnetism collector **23** to be applied to the sensor chip **22**.

(16) As described above, the sensor housing body **20** has the connection parts **11** and **12**. The connection part **11** is formed on the side surface **S1** at a portion in the vicinity of the sensor head **H**, and the connection part **12** is formed on the side surface **S2** at a portion in the vicinity of the sensor head **H**.

(17) FIG. 3 is a schematic plan view of the sensor chip **22**, and FIG. 4 is a cross-sectional view taken along the line A-A in FIG. 3.

(18) As illustrated in FIGS. 3 and 4, the sensor chip **22** has, on its element formation surface, four magnetoresistance effect elements **M1** to **M4** and a cancel coil **C1**. The cancel coil **C1** is covered with an insulating film **24**, and the magnetoresistance effect elements **M1** to **M4** are formed on the insulating film **24**. The magnetoresistance effect elements **M1** to **M4** are covered with an insulating film **25**. The magnetism collector **23** is disposed between the magnetoresistance effect elements **M1**, **M2** and the magnetoresistance effect elements **M3**, **M4** as viewed in the z-direction. With this configuration, a magnetic field in the z-direction collected by the magnetism collector **23** is distributed in the positive x-direction and the negative x-direction on the element formation surface of the sensor chip **22**. As a result, magnetic field components in mutually opposite directions are applied to the magnetoresistance effect elements **M1**, **M2** and magnetoresistance effect elements **M3**, **M4**. The fixed magnetization directions of the magnetoresistance effect elements **M1** to **M4** are made uniform in the positive x-direction or negative x-direction.

(19) The cancel coil **C1** is disposed so as to overlap the magnetoresistance effect elements **M1** to

M4. When a cancel current is made to flow in the cancel coil C1, cancel magnetic fields in mutually opposite directions are applied to the magnetoresistance effect elements M1, M2 and magnetoresistance effect elements M3, M4.

(20) FIG. 5 is a circuit diagram of the magnetic sensor 10.

(21) As illustrated in FIG. 5, the magnetoresistance effect elements M1 to M4 included in the magnetic sensor 10 are bridge-connected, and a differential signal generated thereby is supplied to a differential amplifier 31. The differential amplifier 31 generates a feedback current F1 based on the supplied differential signal. The feedback current F1 flows in the cancel coil C1. This causes the cancel coil C1 to generate a cancel magnetic field so as to make a differential signal component which is an output signal of the magnetic sensor 10 zero. The feedback current F is current-voltage converted by a resistor R, and a detection signal Vout proportional to the feedback current F is generated by a voltage measurement circuit 32 measuring the voltage across the resistor R.

(22) FIG. 6 is a schematic perspective view illustrating a state where the magnetic sensor array 1 according to the present embodiment is deformed.

(23) As illustrated in FIG. 6, by deforming the magnetic sensor array 1 around the rotary shaft 15, an angle θ between two adjacent magnetic sensors 10 can be varied. The long dashed dotted lines Z1 and Z2 illustrated in FIG. 6 denotes the sensitivity axis directions of the respective two adjacent magnetic sensors 10. By thus deforming the magnetic sensor array 1 constituted by the plurality of magnetic sensors 10 into a fan shape, a measurement surface constituted by the sensor heads H of the magnetic sensors 10 can be made into a circular-arc shape. This allows the sensor heads H of the magnetic sensors 10 to be pressed perpendicularly against even a curved surface of an object to be measured. This deformation state of the magnetic sensor array 1 can be maintained by friction force of the rotary shaft 15. To reliably maintain the posture of the magnetic sensor array 1, a screw may be used as the rotary shaft 15 and fixed by a nut. In this case as well, the screw and nut are preferably made of a non-magnetic material.

(24) As described above, in the magnetic sensor array 1 according to the present embodiment, the plurality of magnetic sensors 10 are connected through the rotary shafts 15, thereby allowing the shape of the measurement surface constituted by the plurality of sensor heads to be varied in accordance with the surface shape of an object to be measured.

(25) FIG. 7 is a schematic perspective view illustrating the outer appearance of a magnetic sensor array 2 according to a second embodiment of the present invention.

(26) As illustrated in FIG. 7, the magnetic sensor array 2 according to the second embodiment differs from the magnetic sensor array 1 according to the first embodiment in that connection parts 13 and 14 are formed on the sensor housing bodies 20 of the respective magnetic sensors 10 positioned at both end portions in the x-direction. Other basic configurations are the same as those of the magnetic sensor array 1 according to the first embodiment, so the same reference numerals are given to the same elements, and overlapping description will be omitted.

(27) The connection parts 13 and 14 are formed respectively on side surfaces S3 and S4 of the sensor housing body 20, and the positions thereof in the z-direction are the same as those of the connection parts 11 and 12. The connection part 13 has the same shape as the connection part 11, and the connection part 14 has the same shape as the connection part 12. The side surfaces S3 and S4 each constitute the xz surface and are positioned on mutually opposite sides. With this configuration, a plurality of the magnetic sensor arrays 2 according to the second embodiment can be connected in the y-direction.

(28) FIG. 8 is a schematic perspective view illustrating the outer appearance of a magnetic sensor array 3 according to a third embodiment of the present invention.

(29) As illustrated in FIG. 8, the magnetic sensor array 3 according to the third embodiment has a configuration obtained by connecting the plurality of magnetic sensor arrays 2 according to the second embodiment in the y-direction. The magnetic sensor array 3 in the example illustrated in FIG. 8 has groups G1 to G4 each having the same configuration as the magnetic sensor array 2

according to the second embodiment. Specifically, the connection parts **13** belonging to one (e.g., group **G1**) of the groups adjacent in the y-direction and the connection part **14** belonging to the other one (e.g., group **G2**) thereof are axially supported on each other by rotary shafts **16**, whereby the groups **G1** to **G4** are connected in the y-direction. The axial direction of the rotary shaft **16** is along the x-direction and does not overlap the magnetic sensor **10**. The rotary shaft **16** can be made of the same material as the rotary shaft **15**.

(30) FIG. **9** is a schematic perspective view illustrating a state where the magnetic sensor array **3** according to the present embodiment is deformed. As illustrated in FIG. **9**, by varying the angles of the magnetic sensors **10** constituting the magnetic sensor array **3** around the rotary shafts **15** and **16**, a measurement surface constituted by the sensor heads **H** can be deformed into a spherical shape.

(31) While the preferred embodiment of the present invention has been described, the present invention is not limited to the above embodiment, and various modifications may be made within the scope of the present invention, and all such modifications are included in the present invention.

REFERENCE SIGNS LIST

(32) **1-3** magnetic sensor array **10** magnetic sensor **11-14** connection part **15, 16** rotary shaft **20** sensor housing body **21** substrate **22** sensor chip **23** magnetism collector **24, 25** insulating film **31** differential amplifier **32** voltage measurement circuit **C1** cancel coil **F** feedback current **G1-G4** group **H** sensor head **M1-M4** magnetoresistance effect element **R** resistor **S1-S4** side surface

Claims

1. A magnetic sensor array comprising a plurality of magnetic sensors each elongated in a sensitivity axis direction thereof, wherein each of the magnetic sensors includes a substrate elongated in the sensitivity axis direction, a magnetism collector mounted on the substrate and elongated in the sensitivity axis direction, a sensor chip mounted on the substrate so as to face one end of the magnetism collector in the sensitivity axis direction, and a sensor housing body, wherein the sensor housing body is larger in length in the sensitivity axis direction than the substrate so as to house therein the substrate, the magnetism collector, and the sensor chip, wherein the sensor housing body has a sensor head positioned at another end of the magnetism collector in the sensitivity axis direction, first and second side surfaces parallel to the sensitivity axis direction and positioned on mutually opposite sides, a first connection part formed on the first side surface at a portion in a vicinity of the sensor head, and a pair of second connection parts formed on the second side surface at a portion in a vicinity of the sensor head, and wherein the first connection part formed on one of adjacent magnetic sensors and the pair of second connection parts formed on other one thereof are axially supported on each other such that the first connection part is sandwiched between the pair of second connection parts to achieve mutual connection between the plurality of magnetic sensors.

2. The magnetic sensor array as claimed in claim 1, wherein the plurality of magnetic sensors constitute a plurality of groups including first and second groups, wherein the sensor housing body of at least one of the magnetic sensors belonging to the first group further has a third connection part, wherein the sensor housing body of at least one of the magnetic sensors belonging to the second group further has a fourth connection part, and wherein the first and second groups are mutually connected with the third and fourth connection parts axially supported on each other.

3. The magnetic sensor array as claimed in claim 2, wherein the sensor housing body further has third and fourth side surfaces parallel to the sensitivity axis direction of the magnetic sensor and perpendicular to the first and second side surfaces and are positioned on mutually opposite sides, wherein the third connection part is formed on the third side surface at a portion in a vicinity of the sensor head, and wherein the fourth connection part is formed on the fourth side surface at a portion in a vicinity of the sensor head.

4. The magnetic sensor array as claimed in claim 1, wherein a rotary shaft axially supporting the first connection part and the pair of second connection parts is a screw made of a non-magnetic material.
